

Summer Internship Project Report

Project Title

**Time Series Analysis of Global Temperature Data and K-Means
Clustering on MNIST Handwritten Digits**

Submitted By-

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1 Abstract

This project focuses on the time-series analysis of global temperature data, utilizing historical records from various sources to identify trends and patterns in global warming. The analysis involves data loading, cleaning, transformation, and the application of statistical methods to derive meaningful insights. Key steps include converting date formats, calculating moving averages, filtering data for specific periods, and aggregating data by year and month. Additionally, the project explores an application of K-Means clustering on the MNIST dataset to demonstrate machine learning concepts, evaluating its performance using the F1 score. The primary goal is to provide a comprehensive understanding of global temperature changes over time and to showcase practical data science skills.

2. Introduction

Global temperature data analysis is crucial for understanding climate change and its potential impacts. This project delves into historical global temperature records, specifically focusing on data from sources like GISTEMP (NASA Goddard Institute for Space Studies) and GCAG (UK Met Office HadCRUT5) [2]. The relevance of this project lies in its ability to quantify temperature anomalies and observe long-term warming trends, which are critical for environmental policy-making and scientific research. The technology involved primarily includes Python programming with libraries such as Pandas for data manipulation, NumPy for numerical operations, and Scikit-learn for machine learning tasks. The background material survey indicates that time-series analysis is a fundamental approach to studying climate data, allowing for the identification of seasonality, trends, and irregular components. The procedure used in this project involves a systematic approach to data handling and analysis, from initial data loading to advanced statistical and machine learning techniques. The purpose of this project is to provide practical experience in data science workflows, particularly in the context of environmental data.

During the first two weeks of the internship, training was provided on foundational Data Science skills, which likely included:

- **Python Programming Fundamentals:** Syntax, data structures, control flow, functions.
- **Data Manipulation with Pandas:** DataFrame operations, data loading, cleaning, merging, and aggregation.
- **Numerical Computing with NumPy:** Array operations, mathematical functions.
- **Data Visualization:** Basic plotting techniques using libraries like Matplotlib or Seaborn (though not explicitly shown in the notebook, it's a common initial training topic).
- **Introduction to Machine Learning:** Concepts of supervised and unsupervised learning, basic algorithms like K-Means clustering.
- **Version Control with Git/GitHub:** Basics of repository management, committing, and pushing code.

3. Project Objective

The primary objectives of this project are:

- To load and preprocess global temperature time-series data from multiple sources.
- To perform data cleaning and transformation, including date conversion and handling of missing values.
- To calculate and analyze key statistical measures such as 12-month moving averages to identify underlying trends.
- To aggregate temperature data by different time granularities (e.g., yearly, monthly) to observe patterns.
- To apply unsupervised machine learning techniques, specifically K-Means clustering, to a separate dataset (MNIST) to demonstrate clustering capabilities.
- To evaluate the performance of the clustering model using appropriate metrics like the F1 score.

4. Methodology

The project commenced with loading the monthly_csv.csv dataset into a Pandas DataFrame. The 'Date' column, initially a string, was converted to datetime objects to facilitate time-series operations. The DataFrame was then sorted by date in ascending order. Initial data inspection confirmed the shape of the DataFrame and the absence of missing values, ensuring data quality. New columns for 'Year' and 'Month' were extracted from the 'Date' column to enable time-based aggregations.

A crucial step in the time-series analysis was the calculation of the 12-month moving average of the 'Mean' temperature for each 'Source'. This helped in smoothing out short-term fluctuations and highlighting longer-term trends in temperature anomalies. The data was also filtered to focus on the last 20 years, allowing for a more recent trend analysis.

Further analysis involved grouping the data by 'Source' to calculate the mean 'Mean' temperature for each source, and aggregating by 'Year' to determine the average 'Mean' temperature per year. A pivot table was constructed using the df_last20 DataFrame to display the average 'Mean' temperature for each month across each year, with missing values filled with zero.

In addition to time-series analysis, the project included a demonstration of machine learning using the MNIST dataset. The sklearn.datasets.load_digits function was used to load the dataset, separating features (X) and target labels (y). K-Means clustering was applied with n_clusters=10, corresponding to the 10 digits (0-9). The KMeans model was initialized with random_state=42 and n_init=10 for reproducibility and robustness. The cluster labels (kmeans_labels) were then obtained. To evaluate the clustering performance, the F1 score was calculated. This required mapping each K-Means cluster label to the most frequent true label within that cluster, as K-Means assigns arbitrary labels. The scipy.stats.mode function was used for this mapping, and sklearn.metrics.f1_score was employed to compute the macro-averaged F1 score.

Tools and Methods Used:

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, SciPy
- **Data Handling:** CSV file loading, datetime conversion, data sorting, filtering, grouping, pivot tables.
- **Statistical Analysis:** Moving averages, mean calculations.
- **Machine Learning:** K-Means Clustering.
- **Evaluation Metrics:** F1 Score.

5. Data Analysis and Results

Data Overview and Preprocessing

Upon loading, the `df_temp` DataFrame contained 3288 rows and 3 columns (Source, Date, Mean). After converting the 'Date' column to datetime objects and sorting, no missing values were found. The addition of 'Year' and 'Month' columns facilitated granular analysis.

Source	Date	Mean	Year	Month
GISTEMP	1880-01-06	-0.3000	1880	1
GCAG	1880-01-06	0.0009	1880	1
GCAG	1880-02-06	-0.1229	1880	2
GISTEMP	1880-02-06	-0.2100	1880	2
GCAG	1880-03-06	-0.1357	1880	3

Moving Averages

The 12-month moving average (Moving_Avg) was calculated for each source, providing a smoothed view of temperature trends. The last five rows of the DataFrame with the Moving_Avg column are shown below:

Source	Date	Mean	Year	Month	Moving_Avg
GCAG	2016-10-06	0.7292	2016	10	0.981917

Source	Date	Mean	Year	Month	Moving_Avg
GISTEMP	2016-11-06	0.9300	2016	11	1.017500
GCAG	2016-11-06	0.7504	2016	11	0.963992
GISTEMP	2016-12-06	0.8100	2016	12	0.992500
GCAG	2016-12-06	0.7895	2016	12	0.936292

Descriptive Analysis

Average Mean Temperature by Source:

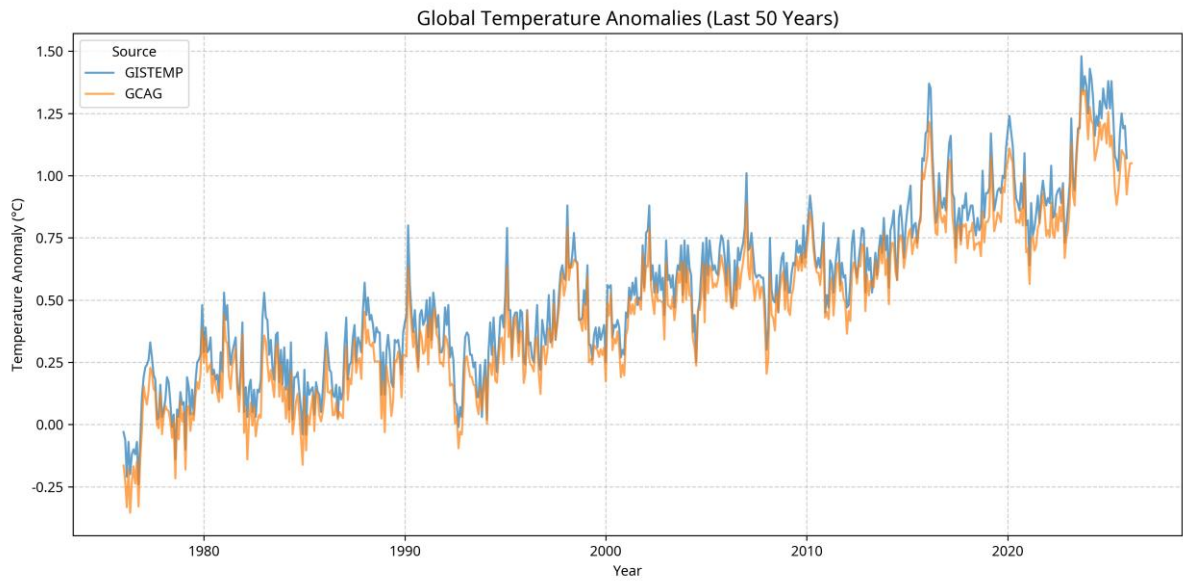
- GCAG: 0.048797
- GISTEMP: 0.024380

Yearly Average Mean Temperature (Last 5 Years):

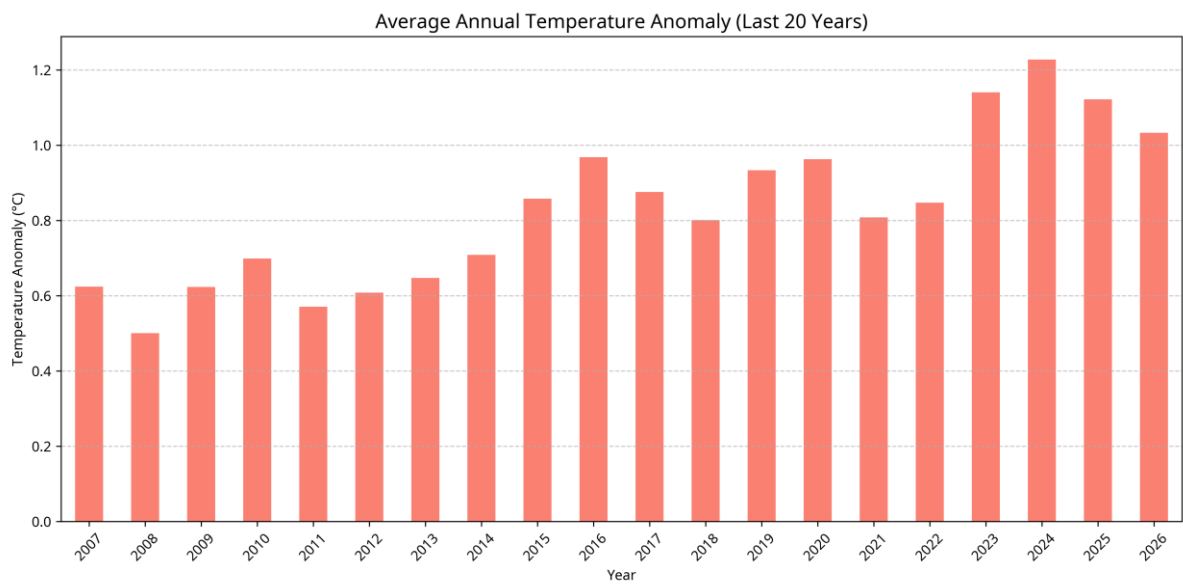
Year	Mean
2012	0.629517
2013	0.661846
2014	0.742075
2015	0.882408
2016	0.964396

This shows a clear increasing trend in average global temperatures over recent years.

Global Temperature Anomalies (Last 50 Years):



Average Annual Temperature Anomaly (Last 20 Years):



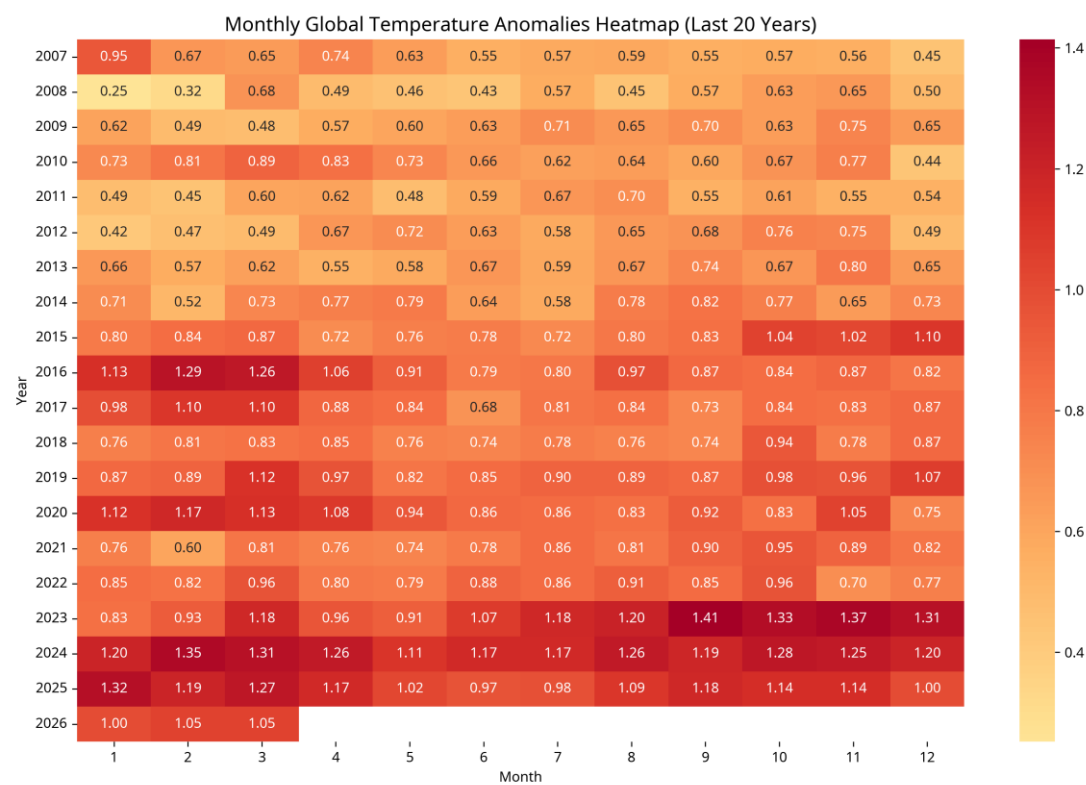
Monthly Average Mean Temperature:

Month	Mean
1	0.021871
2	0.029409

Month	Mean
3	0.045972
4	0.034341
5	0.031106
6	0.022287
7	0.037955
8	0.039577
9	0.050192
10	0.052566
11	0.043543
12	0.030244

October (Month 10) appears to have the highest average mean temperature across all years.

Monthly Global Temperature Anomalies Heatmap (Last 20 Years)



The pivot table below illustrates the average mean temperature for each month across the last 20 years of the dataset, providing a detailed view of monthly temperature variations and trends.

Ye ar	1	2	3	4	5	6	7	8	9	10	11	12
1997	0.35230	0.39940	0.49085	0.41755	0.40675	0.54040	0.43005	0.48955	0.59250	0.64435	0.62325	0.61510
1998	0.60710	0.88300	0.63480	0.68790	0.68895	0.72285	0.72290	0.68080	0.48855	0.48325	0.45660	0.56980
1999	0.49035	0.66665	0.35510	0.39950	0.36930	0.39515	0.40645	0.35475	0.41380	0.40365	0.39440	0.51895
2000	0.30335	0.56195	0.57805	0.61055	0.43285	0.42380	0.40595	0.44285	0.42920	0.30135	0.29785	0.30960
2001	0.45600	0.42730	0.60465	0.54330	0.59485	0.54365	0.58430	0.54660	0.53250	0.51030	0.69400	0.53140
2002	0.72310	0.77230	0.84935	0.57625	0.60780	0.56450	0.62145	0.54295	0.60805	0.52365	0.59090	0.43330

Year	1	2	3	4	5	6	7	8	9	10	11	12
2003	0.71035	0.55630	0.57010	0.55505	0.60770	0.51405	0.54540	0.64010	0.65060	0.73855	0.56420	0.74305
2004	0.59395	0.71660	0.66820	0.60700	0.43210	0.45035	0.37030	0.48205	0.53155	0.63885	0.74005	0.51405
2005	0.67115	0.54715	0.69090	0.71070	0.65365	0.67110	0.65860	0.62735	0.73335	0.74650	0.74400	0.65175
2006	0.51335	0.66105	0.61155	0.50885	0.50995	0.63995	0.58090	0.67155	0.63685	0.68370	0.68245	0.76960
2007	0.92195	0.68045	0.68340	0.74680	0.63950	0.55870	0.57430	0.57545	0.60875	0.58350	0.54360	0.49360
2008	0.25730	0.36870	0.74625	0.50635	0.50555	0.50570	0.58980	0.49970	0.59090	0.66950	0.66515	0.55175
2009	0.60775	0.54895	0.55470	0.63195	0.61840	0.66205	0.68195	0.67990	0.70415	0.64090	0.72080	0.63380
2010	0.71290	0.74400	0.88350	0.84935	0.75410	0.68695	0.67435	0.65475	0.59120	0.66840	0.77910	0.48970
2011	0.49030	0.51380	0.61575	0.65110	0.55205	0.61825	0.67910	0.67300	0.57990	0.64745	0.52285	0.54425
2012	0.44140	0.45675	0.55090	0.71175	0.74105	0.66195	0.62920	0.64970	0.74330	0.74110	0.72935	0.49775
2013	0.63365	0.59285	0.63695	0.54050	0.66205	0.66690	0.62810	0.66025	0.73285	0.68435	0.81965	0.68405
2014	0.71180	0.49500	0.76975	0.79030	0.83190	0.71115	0.63440	0.80930	0.84360	0.81755	0.67975	0.81040
2015	0.81205	0.87215	0.89845	0.75705	0.81920	0.83175	0.75860	0.82730	0.86800	1.02565	1.00275	1.11595
2016	1.11345	1.27105	1.26225	1.08165	0.90190	0.83030	0.84935	0.93990	0.87335	0.80960	0.84020	0.79975

Machine Learning Results (K-Means Clustering)

For the MNIST dataset, the K-Means clustering model was successfully fitted. The shapes of the features ($\underline{X}$) and target labels ($\underline{y}$) were (1797, 64) and (1797,) respectively. After mapping the cluster labels to the true digit labels, the macro-averaged F1 score was calculated.

F1 Score: 0.7894620863890832

This F1 score indicates a reasonably good performance of the K-Means algorithm in grouping similar digits, even without prior knowledge of the true labels, after appropriate label mapping.

6. Conclusion

This internship project successfully demonstrated a comprehensive approach to time-series analysis of global temperature data and an application of K-Means clustering. The analysis of global temperature data revealed a clear upward trend in average temperatures, particularly in recent decades, as evidenced by the yearly average mean temperatures and the 12-month moving averages. This finding aligns with broader scientific consensus on global warming. The project also highlighted monthly temperature variations, with October showing the highest average mean temperature across the dataset.

The K-Means clustering on the MNIST dataset, despite being an unsupervised learning technique, achieved a respectable F1 score of approximately 0.789. This demonstrates the effectiveness of clustering algorithms in identifying inherent structures within data, even when ground truth labels are not used during training. The methodology employed, encompassing data preprocessing, statistical analysis, and machine learning, provides a solid foundation for further exploration in climate science and data analytics.

Recommendations for Future Work:

- **Advanced Time-Series Models:** Implement more sophisticated time-series forecasting models such as ARIMA, SARIMA, or Prophet to predict future temperature trends.
- **Causal Analysis:** Investigate potential causal factors contributing to temperature changes, such as greenhouse gas emissions, solar activity, and volcanic eruptions.
- **Geospatial Analysis:** Incorporate geographical data to analyze regional temperature variations and identify hotspots.
- **Deep Learning for Time Series:** Explore recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) networks for more complex pattern recognition in temperature data.
- **Interactive Visualizations:** Develop interactive dashboards to allow users to explore temperature trends and anomalies dynamically.

7. APPENDICES

1. References

[1] IDEAS Technology Innovation Hub. (n.d.). *Learning & Development*. Retrieved from <https://www.ideas-tih.org/skilldevelopment> [2] DataHub. (2026, May 14). *Global Temperature Time Series*. Retrieved from <https://datahub.io/core/global-temp>

2. Survey Questionnaire (if any)

Not applicable for this project.

3. Github link for the codes developed (if any)

<https://github.com/Biswarup760/Summe-Internship-Biswarup-Goswami->

Thank you
